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United States Patent
Kind Code
Date of Patent
Inventor(s)

12414980
B2
September 16, 2025
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Pharmaceutical composition comprising RUNX3 protein and CDK4 inhibitor or mTOR inhibitor cotreatment for prevention or treatment of cancer

Abstract

The present invention relates to a pharmaceutical composition for prevention or treatment of cancer, comprising a Runx3 (Runt-related transcription factor 3) protein, a polynucleotide encoding thereof, a vector carrying the polynucleotide or a virus or cell transformed with the vector; and a CDK4 inhibitor or an mTOR inhibitor as an active ingredient. It was confirmed that the binding of Runx3 and BRD2 was maintained for up to 8 hours when the CDK4 inhibitor (PD0332991) was present at the concentration of 11 nM known to be non-toxic to normal cells. It was also confirmed that the binding of Runx3 and BRD2 was maintained for up to 8 hours when the mTOR inhibitor (rapamycin) was present at the concentration of 100 nM known to be non-cytotoxic to normal cells. In addition, it was confirmed that the cancer apoptotic effect was significantly increased when the CDK4 inhibitor or mTOR inhibitor was administered simultaneously with Runx3 than when Runx3 was administered alone to Runx3 deficient cancer cells. Therefore, the CDK4 inhibitor or mTOR inhibitor can be effectively used for preventing or treating various cancers by administering the same in combination with Runx3.

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Appl. No.: 17/239433

Filed: April 23, 2021

Prior Publication Data

Document Identifier

US 20210330742 A1

Publication Date

Oct. 28, 2021

Foreign Application Priority Data

KR	10-2020-0049341	Apr. 23, 2020
KR	10-2020-0184526	Dec. 28, 2020

Publication Classification

Int. Cl.: **A61K38/17** (20060101); **A61K31/436** (20060101); **A61K31/519** (20060101); **A61K47/46** (20060101); **A61P35/00** (20060101); **C07K14/47** (20060101); **C12N7/00** (20060101)

U.S. Cl.:

CPC **A61K38/17** (20130101); **A61K31/436** (20130101); **A61K31/519** (20130101); **A61K38/1709** (20130101); **A61K47/46** (20130101); **A61P35/00** (20180101); **C07K14/4702** (20130101); **C12N7/00** (20130101); C12N2750/14143 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2008/0261883	12/2007	Bae et al.	N/A	N/A
2012/0128642	12/2011	Teumer	435/7.1	G01N 33/6881
2015/0336967	12/2014	Czardybon	540/603	C07D 403/12
2018/0002393	12/2017	Bancel et al.	N/A	N/A
2021/0330740	12/2020	Bae et al.	N/A	N/A
2022/0088126	12/2021	Bae et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-2002-0063387	12/2001	KR	N/A
10-1994957	12/2018	KR	N/A
WO 2013/151672	12/2012	WO	N/A

OTHER PUBLICATIONS

Berns, "Cancer: The blind spot of p53," *Nature* 468.7323: 519-520, Nov. 2010. cited by applicant
Feldser et al., "Stage-specific sensitivity to p53 restoration during lung cancer progression," *Nature* 468.7323: 572-575, Nov. 2010. cited by applicant
Ito et al., "The RUNX family: developmental regulators in cancer," *Nature Reviews Cancer* 15: 81-

95, Jan. 2015. cited by applicant

Junttila et al., "Selective activation of p53-mediated tumour suppression in high-grade tumours," *Nature* 468.7323: 567-571, Nov. 2010. cited by applicant

Lee et al., "Runx3 is required for the differentiation of lung epithelial cells and suppression of lung cancer," *Oncogene* 29: 3349-3361, Mar. 2010. cited by applicant

Lee et al., "Runx3 Inactivation is a Crucial Early Event in the Development of Lung Adenocarcinoma," *Cancer Cell* 24: 603-616, Nov. 2013. cited by applicant

Lee et al., "RUNX3 protects against oncogenic KRAS," *Cancer Discovery* 4.1: 14, Jan. 2014. cited by applicant

Lee et al., "RUNX3 regulates cell cycle-dependent chromatin dynamics by functioning as a pioneer factor of the restriction-point," *Nat Commun.* 10: 1897, 2019 (17 pages). cited by applicant

"Study on the decision making mechanism for cell division," Final (Result) Report of Research Fellow for Academic Successor Generation Project, 2017 (w/English translation of summary of research results). cited by applicant

Walter et al., "RB constrains lineage fidelity and multiple stages of tumour progression and metastasis," *Nature* 569.7756: 423-427, May 2019 (w/supplemental information). cited by applicant

Wang et al., "Comparison of Adenoviral and Adeno-Associated Viral Vectors for Pancreatic Gene Delivery In Vivo," *Hum Gene Ther.* 15.4: 405-413, Apr. 2004. cited by applicant

Wang et al., "Targeting adeno-associated virus and adenoviral gene therapy for hepatocellular carcinoma," *World J Gastroenterol.* 22.1: 326-337, Jan. 2016. cited by applicant

El-Aneed, Anas, "An overview of current delivery systems in cancer gene therapy," *Journal of Controlled Release*, vol. 94, pp. 1-14, 2004. cited by applicant

Strausberg et al., "Generation and initial analysis of more than 15,000 full-length human and mouse cDNA sequences," *PNAS*, vol. 99, No. 26, pp. 16899-16903, 2002. cited by applicant

Mukhopadhyay et al., "The Enigma of Rapamycin Dosage," *Molecular Cancer Therapeutics*, 15(3): 347-353, 2016. cited by applicant

Xue et al., "Therapeutic evaluation of palbociclib and its compatibility with other chemotherapies for primary and recurrent nasopharyngeal carcinoma," *Journal of Experimental & Clinical Cancer Research*, 39: Article 262, 2020 (24 pages). cited by applicant

Zhang et al., "PTHrP prevents chondrocyte premature hypertrophy by inducing cyclin-D1-dependent Runx2 and Runx3 phosphorylation, ubiquitylation and proteasomal degradation," *J Cell Sci*, vol. 122 (Pt. 9), pp. 1382-1389, 2009. cited by applicant

Kemp et al., "Elimination of background recombination: somatic induction of Cre by combined transcriptional regulation and hormone binding affinity," *Nucleic Acids Research* 32.11: e92, Jul. 2004 (7 pages). cited by applicant

Kudo et al., "Oncogenic Role of RUNX3 in Head and Neck Cancer," *J Cell. Biol.* 112: 387-393, 2011. cited by applicant

Lee et al., "RUNX3 functions as an oncogene in ovarian cancer," *Gynecologic Oncology* 122: 410-417, May 2011. cited by applicant

Stolze et al., "Comparative analysis of KRAS codon 12, 13, 18,61, and 117 mutations using human MCFIOA isogenic cell lines," *Scientific Reports* 5: 8535, Feb. 2015 (9 pages). cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority from Korean Patent Application No. 10-2020-0049341, filed on Apr. 23, 2020, and Korean Patent Application No. 10-2020-0184526, filed on Dec. 28, 2020, the contents of each of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a pharmaceutical composition comprising a Runx3 protein and a CDK4 inhibitor or an mTOR inhibitor as an active ingredient.

2. Description of the Related Art

(3) Research on the development of targeted cancer therapy is focused on strategies to control cancer cells by inhibiting the function of an oncogene or activating the function of a tumor suppressor gene. Abnormal activation of K-Ras function by mutation of K-Ras among the oncogenes is known as one of the major causes of human cancer. The mutation of K-Ras is also observed in lung cancer, and it is known that the mutation of K-Ras is observed in about 35% of lung adenocarcinoma. Thus, in order to treat cancer caused by the activation of K-Ras function, studies have been conducted on a method of treating cancer by inhibiting the function of K-Ras. However, a strategy that directly inhibits the function of K-Ras has not been developed as a successful anticancer drug because it causes serious damage to normal cells. Therefore, instead of suppressing the function of an oncogene, a strategy of activating the inhibited function of a tumor suppressor gene is receiving attention. Therefore, instead of a strategy for inhibiting the function of an oncogene, a strategy for activating the inhibited function of a tumor suppressor gene is attracting attention.

(4) The said tumor suppressor gene refers to a nucleotide sequence that can be expressed in a target cell to suppress a tumor phenotype or induce apoptosis. The tumor suppressor genes identified so far include SPD-1, VHL, MMAC1, DCC, p53, NF1, WT1, Rb, BRCA1 and BRCA2. Among them, it has been reported that p53 or Rb gene is frequently inhibited in its function in K-Ras mutant cancers. Whether it is possible to treat K-Ras mutant cancer through the repair of the suppressor gene has become a subject of great interest in the field of anticancer agent development research. Accordingly, there have been attempts to treat K-Ras mutant lung adenocarcinoma by recovering the function of p53 gene, which is a representative tumor suppressor gene, but it was not successful because early lung adenocarcinoma was not cured (Feldser, D. M. et al., Nature, 468: 572-575, 2010, Junttila, M. R. et al., Nature, 468: 567-571, 2010). In addition, it was found that K-Ras mutant lung cancer could not be cured through the recovery of Rb gene function (Walter, D. M. et al. Nature 2019). The above results indicate that even if the function of the tumor suppressor gene is simply restored, the therapeutic effect on the already-onset cancer does not appear, because the early stage cancer rapidly develops into a malignant cancer (Berns A., Nature, 468:519-520, 2010). There have been no reports of successful treatment of K-Ras mutant lung cancer through the activation of a tumor suppressor gene.

(5) It has been reported that the function of Runx3 gene as a tumor suppressor gene is inhibited in K-Ras mutant cancers (RUNX3 Protects against Oncogenic KRAS. (2013). Cancer Discovery, 4(1), 14-14), and that the activity of Runx3 gene is inhibited in lung adenocarcinoma caused by the mutation of K-Ras (Lee, K. S., Lee, Y. S., Lee, J. M., Ito, K., Cinghu, S., Kim, J. H., Bae, S. C. Oncogene, 29(23): 3349-61, 2010).

(6) Runx3, a transcription factor that binds to DNA, plays a crucial role in lineage determination (Ito, Y., Bae, S. C. & Chuang, L. S. The RUNX family: developmental regulators in cancer. Nat. Rev. Cancer 15, 81-95 (2015)). Deletion of Runx3 in the mouse lung leads to the development of lung adenomas and accelerates the progression to adenocarcinoma (ADCs) induced by K-Ras.

(7) Thus, the present inventors have completed the present invention by confirming that the cancer apoptotic effect was significantly increased when the CDK4 inhibitor or mTOR inhibitor was

administered simultaneously with Runx3 protein compared to when Runx3 was introduced alone into Runx3 deficient cancer cells since the CDK4 inhibitor and mTOR inhibitor increased the time for maintaining the Runx3-BRD2 complex.

SUMMARY OF THE INVENTION

(8) It is an object of the present invention to provide a pharmaceutical composition for prevention or treatment of cancer.

(9) To achieve the above object, the present invention provides a pharmaceutical composition for prevention or treatment of cancer, comprising a modified protein in which the 356^{sup.th} serine of Runx3 (Runt-related transcription factor 3) protein is substituted with a hydrophobic amino acid, a polynucleotide encoding thereof, a vector carrying the polynucleotide, or a virus or cell transformed with the vector as an active ingredient.

(10) The present invention also provides a pharmaceutical composition for prevention or treatment of cancer, comprising a Runx3 (Runt-related transcription factor 3) protein, a polynucleotide encoding thereof, a vector carrying the polynucleotide or a virus or cell transformed with the vector; and a CDK4 inhibitor as an active ingredient.

(11) In addition, the present invention provides a pharmaceutical composition for prevention or treatment of cancer, comprising a Runx3 (Runt-related transcription factor 3) protein, a polynucleotide encoding thereof, a vector carrying the polynucleotide or a virus or cell transformed with the vector; and an mTOR inhibitor as an active ingredient.

Advantageous Effect

(12) The present invention relates to a pharmaceutical composition for prevention or treatment of cancer, comprising a Runx3 (Runt-related transcription factor 3) protein, a polynucleotide encoding thereof, a vector carrying the polynucleotide or a virus or cell transformed with the vector; and a CDK4 inhibitor or an mTOR inhibitor as an active ingredient. It was confirmed that the binding of Runx3 and BRD2 was maintained for up to 8 hours when the CDK4 inhibitor (PD0332991) was present at the concentration of 11 nM known to be non-toxic to normal cells. It was also confirmed that the binding of Runx3 and BRD2 was maintained for up to 8 hours when the mTOR inhibitor (rapamycin) was present at the concentration of 100 nM known to be non-toxic to normal cells. In addition, it was confirmed that the cancer apoptotic effect was significantly increased when the CDK4 inhibitor or mTOR inhibitor was administered simultaneously with Runx3 than when Runx3 was administered alone to Runx3 deficient cancer cells. Therefore, the CDK4 inhibitor or mTOR inhibitor can be effectively used for preventing or treating various cancers by administering the same in combination with Runx3.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1*a* is a schematic diagram illustrating the BRD2 structure and the interacting proteins, confirming that BD1 interacted with the RUNX3 acetylated at Lys-94 and Lys-171, BD2 interacted with the acetylated histones H4K4-ac, H4K12-ac and H3K14-ac, and C-terminal region interacted with TFIID and SWI/SNF complex.

(2) FIGS. 1*b* and 1*c* are diagrams illustrating the level of protein that formed a complex with Runx3 in HEK293 cells induced mitogenic stimulation by stimulation with 10% serum after serum depletion for 24 hours, measured by immunoprecipitation (IP) and immunoblotting (IB).

(3) FIG. 2*a* is a diagram confirming that E2F1 was combined with RUNX3 by measuring the time-dependent formation of BRD2-RUNX3, E2F1-RUNX3, CDK4-RUNX3, Cyclin D1-RUNX3, HDAC4-RUNX3, p16INK4a-CDK4, p21-CDK4, Cyclin D1-CDK4 and HDAC4-CDK4 complexes by immunoprecipitation (IP) and immunoblotting (IB), and the time-dependent phosphorylation of pRB (at Ser-795) and ERK1/2 by immunoblotting (IB).

- (4) FIG. 2b is a diagram confirming the physical binding between RUNX3 and CDK4 through proximity ligation analysis (PLA) 2 hours after serum stimulation.
- (5) FIG. 2c is a diagram illustrating the time-dependent formation of BRD2-RUNX3, CDK4-RUNX3, HDAC4-RUNX3 and Cyclin D1-RUNX3 complexes in HEK293 cells treated with control or CDK4-specific siRNA (si-con or si-CDK4), and stimulated with serum after serum depletion for 24 hours, measured by immunoprecipitation (IP) and immunoblotting (IB). The time-dependent expression of ARF was measured by immunoblotting (IB), and it was confirmed that the knockdown of CDK4 significantly reduced the binding of RUNX3 to Cyclin D1 and HDAC4.
- (6) FIG. 3a is a diagram illustrating the time-dependent interaction of BRD2-RUNX3 and p300-RUNX3 as well as the phosphorylation of ERK1/2, measured by immunoprecipitation (IP) and immunoblotting (IB), confirming that the binding of BRD2-RUNX3 and p300-RUNX3 was eliminated by the treatment of the MEK1 inhibitor.
- (7) FIG. 3b is a diagram illustrating the time-dependent formation of BRD2-RUNX3 and CDK4-RUNX3 complexes and the phosphorylation of RUNX3 at Ser-356 and pRB at Ser-795, measured by immunoprecipitation (IP) and immunoblotting (IB), confirming that the knockdown of Cyclin D1 effectively reduced the phosphorylation of pRB (S795), but did not affect the phosphorylation of RUNX3 (S356).
- (8) FIG. 3c is a diagram illustrating the time-dependent formation of JNK-CDK4 complex and the phosphorylation of RUNX3 at Ser-356 and CDK4 at Thr-172, measured by immunoprecipitation (IP) and immunoblotting (IB), confirming that JNK phosphorylated T172 of CDK4 2 hours after serum stimulation.
- (9) FIG. 3d is a diagram illustrating the time-dependent formation of BRD2-RUNX3 and RUNX3-CDK4 complexes in Ser-356 and the phosphorylation of RUNX3 measured by immunoprecipitation (IP) and immunoblotting (IB), and the time-dependent expression of ARF measured by immunoblotting (IB), confirming that the pharmacological inhibition of JNK activity remarkably reduced the phosphorylation of RUNX3 Ser-356 and maintained Rpa-RX3-AC for up to 8 hours.
- (10) FIG. 3e is a diagram illustrating the time-dependent formation of BRD2-RUNX3 and CDK4-RUNX3 complexes in Ser-356 and the phosphorylation of RUNX3 measured by immunoprecipitation (IP) and immunoblotting (IB), and the time-dependent expression of ARF measured by immunoblotting (IB), confirming that the knockdown of JNK by siRNA also reduced the phosphorylation of RUNX3 Ser356 and maintained Rpa-RX3-AC.
- (11) FIG. 3f is a diagram illustrating the time-dependent formation of BRD2-RUNX3 and CDK4-RUNX3 complexes in Ser-356 and the phosphorylation of RUNX3 measured by immunoprecipitation (IP) and immunoblotting (IB), and the time-dependent phosphorylation of pRB and the expression of ARF measured by immunoblotting (IB), confirming that the over-expressed CDK4-WT phosphorylated Myc-RUNX3 2-4 hours after serum stimulation, but Flag-CDK4-T172A (a mutant made to prevent phosphorylation by JNK) did not phosphorylate RUNX3 Ser356.
- (12) FIG. 3g is a diagram illustrating the time-dependent formation of BRD2-RUNX3 complex measured by immunoprecipitation (IP) and immunoblotting (IB), and the time-dependent expression of ARF measured by immunoblotting (IB), confirming that the knockdown of a PI3K catalytic subunit (PIK3CA, encoding p110a) reduced the level of Cyclin D1, maintained Rpa-RX3-AC, and extended the expression of ARF up to 8 hours.
- (13) FIG. 3h is a diagram illustrating the time-dependent formation of BRD2-RUNX3 complex measured by immunoprecipitation (IP) and immunoblotting (IB), and the time-dependent expression of ARF measured by immunoblotting (IB), confirming that the ribosomal protein S6 kinase beta-1 (S6K1) phosphorylated by mTOR signaling was used for control.
- (14) FIG. 4 is a schematic diagram illustrating the role of the RAS pathway in regulating the R-point transition, confirming that the RAS-RAF-MEK pathway promoted the formation of Rpa-

RX3-AC and inhibited the formation of PRC1-Cyclin D1-HDAC4 complex, the RAS-RAC-JNK pathway activated CDK4 in Rpa-RX3-AC, and the RAS-PI3K pathway promoted the formation of Rpa-RX3-TR by contributing to the translation of Cyclin D1.

(15) FIG. 5 is a diagram confirming that the binding of Runx3 and BRD2 was maintained for up to 8 hours when the CDK4 inhibitor (PD0332991) was present at the concentration of 11 nM known to be non-toxic to normal cells.

(16) FIG. 6 is a diagram confirming that the binding of Runx3 and BRD2 was maintained for up to 8 hours when the mTOR inhibitor (rapamycin) was present at the concentration of 100 nM known to be non-cytotoxic to normal cells.

(17) FIG. 7a is a diagram confirming the cancer apoptotic effect when Runx3 was introduced into Runx3 deficient H460 lung cancer cell line and when Runx3 and the CDK4 inhibitor were simultaneously introduced in the presence of the CDK4 inhibitor (PD0332991) at the concentration of 11 nM known to be non-toxic to normal cells, and the cancer apoptotic effect when Runx3 was introduced into Runx3 deficient cancer cells and when Runx3 and the CDK4 inhibitor were simultaneously introduced in the presence of the mTOR inhibitor (rapamycin) at the concentration of 100 nM known to be non-cytotoxic to normal cells.

(18) FIG. 7b is a diagram confirming the cancer apoptotic effect when Runx3 was introduced into Runx3 deficient MKN28 stomach cancer cell line and when Runx3 and the CDK4 inhibitor were simultaneously introduced in the presence of the CDK4 inhibitor (PD0332991) at the concentration of 11 nM known to be non-toxic to normal cells, and the cancer apoptotic effect when Runx3 was introduced into Runx3 deficient cancer cells and when Runx3 and the CDK4 inhibitor were simultaneously introduced in the presence of the mTOR inhibitor (rapamycin) at the concentration of 100 nM known to be non-cytotoxic to normal cells.

(19) FIG. 7c is a diagram confirming the cancer apoptotic effect when Runx3 was introduced into Runx3 deficient PANC1 pancreatic cancer cell line and when Runx3 and the CDK4 inhibitor were simultaneously introduced in the presence of the CDK4 inhibitor (PD0332991) at the concentration of 11 nM known to be non-toxic to normal cells, and the cancer apoptotic effect when Runx3 was introduced into Runx3 deficient cancer cells and when Runx3 and the CDK4 inhibitor were simultaneously introduced in the presence of the mTOR inhibitor (rapamycin) at the concentration of 100 nM known to be non-cytotoxic to normal cells.

(20) FIG. 8 is a diagram confirming that the cancer apoptotic effect was significantly increased in H460 lung cancer cell line, MKN28 stomach cancer cell line and PANC1 pancreatic cancer cell line when the adenovirus expressing Runx3 protein and the CDK4 inhibitor or the mTOR inhibitor were administered simultaneously than when the adenovirus expressing Runx3 protein was administered alone.

SEQUENCE

(21) The Sequence Listing is submitted as an ASCII text file in the form of the file name "Sequence.txt" (~13 kb), which was created on Apr. 22, 2021, and which is incorporated by reference herein.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(22) Hereinafter, the present invention is described in detail.

(23) The present invention provides a pharmaceutical composition for prevention or treatment of cancer, comprising a Runx3 (Runt-related transcription factor 3) protein, a polynucleotide encoding thereof, a vector carrying the polynucleotide or a virus or cell transformed with the vector; and a CDK4 inhibitor as an active ingredient.

(24) The CDK4 inhibitor can be any one selected from the group consisting of Abemaciclib, Palbociclib and Ribociclib, but not always limited thereto.

(25) The CDK4 inhibitor is represented by the following formula 1.

(26) ##STR00001##

(27) The present invention also provides a pharmaceutical composition for prevention or treatment

of cancer, comprising a Runx3 (Runt-related transcription factor 3) protein, a polynucleotide encoding thereof, a vector carrying the polynucleotide or a virus or cell transformed with the vector; and an mTOR inhibitor as an active ingredient.

(28) The mTOR inhibitor can be any one selected from the group consisting of rapamycin, Ridaforolimus, Everolimus and Temsirolimus, but not always limited thereto.

(29) The mTOR inhibitor is represented by the following formula 2.

(30) ##STR00002##

(31) The cancer is solid cancer.

(32) The solid cancer can be one or more selected from the group consisting of lung cancer, pancreatic cancer, liver cancer and stomach cancer, but not always limited thereto.

(33) Runx3 (Runt-related transcription factor 3) gene is one of the Runt family genes consisting of Runx1, Runx2 and Runx3. The Runt family genes play an important role in normal development and oncogenesis, and they function as transcriptional regulators of the Smad family, a downstream factor that mediates TGF- β and its signaling. Runx1 plays an important role in mammalian hematopoiesis, Runx2 plays an important role in bone formation, and Runx3 is mainly expressed in granular gastric mucosal cells, and plays a role in inhibiting cell differentiation of gastric epithelium. These three genes are located at loci of chromosomes 1p, 6p and 21q, of which Runx3 gene is located at 1p36. 11-1p36. 13. The Runx3 locus is one of the sites that are lost in a variety of cancers or affected by hemizygous defects. In addition, Runx3 has been found to be inactivated in various types of cancer, and it is gaining spotlight as a new target for the development of anticancer agents. As such, Runx3 is known to act as a tumor suppressor gene that suppresses the formation of cancer, and plays an important role in the restriction-point, which determines the fate of cell division and death, and induces cell division or apoptosis depending on the situation (Lee et al., Nat Commun. 2019; 10(1): RUNX3 regulates cell cycle-dependent chromatin dynamics by functioning as a pioneer factor of the restriction-point). When a K-Ras oncogene mutation occurs in lung epithelial cells, Runx3 kills cancer cells by contributing to determining apoptosis fate at the restriction-point (Lee et al., Nat Commun. 2019; 10(1)).

(34) A Runx3 protein refers to a Runt-related transcription factor 3 related to the Runt family expressed by the Runx3 gene.

(35) BRD2 (bromodomain-containing protein 2) is a factor that acts as a signaling mediator in the nucleus. It is widely expressed in mammalian cells, and plays an important role in cell cycle regulation and transcriptional regulation.

(36) The said BRD2 binds to the acetylated Runx3 to form Runx3-Brd2 complex.

(37) The bromodomain 1 (BD1) of the BRD2 binds to the lysine residues 94 and 171 of Runx3.

(38) The bromodomain 2 (BD2) of the BRD2 binds to the lysine residue 5 of the acetylated histone 4 of Runx3, the lysine residue 12 of histone 4, and the lysine residue 14 of histone 3.

(39) The CDK4 inhibitor or mTOR inhibitor increases the time for maintaining the Runx3-Brd2 complex.

(40) The Runx3-Brd2 complex is formed by mitogenic stimulation, and contributes to the determination of restriction point (R-point).

(41) The restriction point (R-point) is a step in the cell division process that allows cells to make their own decisions to continue life or die. Normal cells make decisions about life by themselves and divide, but abnormal cells such as cancer cells decide to kill themselves at this R-point, thereby removing mutant cells from the body and maintaining normal cells.

(42) The Runx3 protein can be composed of the amino acid sequence represented by SEQ. ID. NO: 11 or SEQ. ID. NO: 12.

(43) The Runx3 protein can be derived from humans or animals.

(44) The Runx3 protein can be synthesized by the conventional chemical synthesis method in the art (W.H. Freeman and Co., Proteins; structures and molecular principles, 1983), or can be prepared by the conventional genetic engineering method (Maniatis et al., Molecular Cloning: A laboratory

Manual, Cold Spring Harbor laboratory, 1982; Sambrook et al., Molecular Cloning: A Laboratory Manual et al.).

(45) The Runx3 protein can be a variant of an amino acid sequence having a different sequence by deletion, insertion or substitution of amino acid residues, or a combination thereof within a range that does not affect the function of the protein. Amino acid exchanges in proteins that do not totally alter the activity of the molecule are informed in the art. In some cases, the amino acid can be modified by phosphorylation, sulfation, acrylation, glycosylation, methylation or farnesylation. Accordingly, the present invention can include a peptide having an amino acid sequence substantially identical to that of a protein composed of the amino acid sequence represented by SEQ. ID. NO: 1 or SEQ. ID. NO: 2, and variants or fragments thereof. The substantially identical protein can have homology to the protein of the present invention by 80% or more, particularly 90% or more, and more particularly 95% or more.

(46) The polynucleotide encoding the Runx3 protein can be derived from humans or animals.

(47) The polynucleotide encoding the Runx3 protein can be composed of the nucleotide sequence represented by SEQ. ID. NO: 13 or SEQ. ID. NO: 14.

(48) The vector including the polynucleotide encoding the Runx3 protein can be linear DNA or plasmid DNA.

(49) The vector refers to a transport mediator for introducing the polynucleotide encoding the Runx3 protein of the present invention into a subject to be treated, and can include a promoter suitable for expression in a subject to be treated, an enhancer, and a polynucleotide encoding the Runx3 protein, a transcription termination site, and the like. The promoter can be a specific organ and tissue specific promoter, and can include a replication origin so as to proliferate in the organ and tissue.

(50) The virus transformed by the vector can be any one selected from the group consisting of retrovirus, adenovirus, herpes simplex virus and lentivirus, but not always limited thereto.

(51) In the case of the vector containing the polynucleotide, it is preferably to contain 0.05 to 500 mg, and more preferably to contain 0.1 to 300 mg. In the case of the recombinant virus containing the polynucleotide encoding Runx3 protein, it is preferably to contain 10.sup.3 to 10.sup.12 IU (10 to 10.sup.10 PFU), and more preferably to contain 10.sup.5 to 10.sup.10 IU.

(52) The recombinant virus is preferably adenovirus. Adeno-associated virus (AAV) is unsuitable as a delivery vehicle for cancer treatment because its gene expression rate or expression speed is lower than that of adenovirus. Adenovirus is suitable for the delivery of the modified protein according to the present invention to the human body because the transferred gene is expressed in adenovirus more than 3 weeks faster than in adeno-associated virus (HUMAN GENE THERAPY 15:405-413.), and the phenomenon of lowering the gene transfer efficiency due to the immune response is less in the adenovirus than in the adeno-associated virus (World J Gastroenterol. 2016 Jan. 7; 22(1):326-37.).

(53) The number of viruses for treatment can be represented by the number of viral particles including the vector genome or the number of infectable viruses. That is, since about 1% of the virus particles are the effective number of viruses that can actually be infected, IU (infection unit) or PFU (plaque forming unit) is used to indicate this.

(54) The cell transformed by the vector can be bacteria.

(55) The bacterium can be non-pathogenic or non-toxic, and can be *Listeria*, *Shigella*, *Salmonella*, or *E. coli*. By introducing the vector into bacteria, DNA of a gene included in the vector can be mass-replicated or proteins can be mass-produced.

(56) The vector according to the present invention can be introduced into cells using a method known in the art. For example, transient transfection, microinjection, transduction, cell fusion, calcium phosphate precipitation, liposome-mediated transfection, DEAE dextran-mediated transfection, polybrene-mediated transfection, electroporation, gene gun, and other known methods for introducing nucleic acids into cells can be used to introduce the vector into cells, but not always

limited thereto (Wu et al., J. Bio. Chem., 267:963-967, 1992; Wu and Wu, J. Bio. Chem., 263:14621-14624, 1988).

(57) In the case of the cells transformed with the vector containing the polynucleotide, it is preferably to contain 10.sup.3 to 10.sup.8 cells, and more preferably to contain 10.sup.4 to 10.sup.7 cells.

(58) In preferred embodiments of the present invention, the present inventors confirmed that Runx3-BRD2 complex was formed 1 to 2 hours after serum stimulation in which BD1 of BRD2 was bound to the lysine residues 94 and 171 of the acetylated Runx3, BD2 was bound to the lysine residue 5 (H4K5-ac) of the acetylated histone 4, the lysine residue 12 (H4K12-ac) of histone 4 and the lysine residue 14 (H3K14-ac) of histone 3, and the binding of Runx3 and BRD2 was maintained for up to 8 hours in the presence of the CDK4 inhibitor (PD0332991) at the concentration of 11 nM known to be non-toxic to normal cells or the mTOR inhibitor (rapamycin) at the concentration of 100 nM known to be non-cytotoxic to normal cells. The present inventors also confirmed that the cancer apoptotic effect was significantly increased when the CDK4 inhibitor or mTOR inhibitor was administered simultaneously with Runx3 than when Runx3 was administered alone to Runx3 deficient cancer cells.

(59) Therefore, the CDK4 inhibitor or mTOR inhibitor of the present invention can be effectively used for preventing or treating cancer by administering the same in combination with Runx3 protein.

(60) Hereinafter, the present invention will be described in detail by the following examples and experimental examples.

(61) However, the following examples and experimental examples are only for illustrating the present invention, and the contents of the present invention are not limited thereto.

Experimental Methods

(62) 1. Cell Line Preparation

(63) HEK293 cells (ATCC, Manassas, VA, USA) were maintained in DMEM medium (Gibco BRL, Thermo Fisher Scientific, MA, USA, MA) supplemented with 10% fetal bovine serum (Gibco BRL) and 1% penicillin/streptomycin (Invitrogen, Carlsbad, CA, USA).

(64) H460 cells (ATCC, Manassas, VA, USA) and H460 stable cells were maintained in RPMI 1640 medium (Gibco BRL) supplemented with 10% fetal bovine serum (Gibco BRL) and 1% penicillin/streptomycin (Invitrogen). All the cell lines were cultured in a 37° C., 5% CO.sub.2 incubator.

(65) 2. Antibody

(66) The antibody targeting RUNX3 (5G4) (Cat #ab40278) was obtained from Abcam (Cambridge, UK), and the antibody was diluted 1:3000. BRD2 (M01; 1:1000; Cat #H00006046-M01, Abnova, Taipei City, Taiwan) was used for immunoblotting and immunoprecipitation.

(67) 3. DNA Transfection, Immunoprecipitation (IP) and Immunoblotting (IB)

(68) Transient transfection was performed in all cell lines using lipofectamine plus reagent and lipofectamine (Invitrogen). Cell lysates were incubated with an appropriate monoclonal or polyclonal antibody (2 µg of antibody/500 µg of lysate sample) at 4° C. for 3 hours, followed by incubation with protein G-Sepharose beads (Amersham Pharmacia Biotech, Piscataway, NJ, USA). For the detection of endogenous proteins at 4° C. for 1 hour, the lysate was incubated with an appropriate monoclonal or polyclonal antibody (1:1000~1:3000) at 4° C. for 6 to 12 hours, and then protein G-Sepharose beads (Amersham Pharmacia Biotech) were heated at 4° C. for 3 hours. The immune precipitate was digested on an SDS-polyacrylamide gel electrophoresis (SDS-PAGE) gel and transferred to a PVDF membrane (Millipore, Billerica, MA, USA). The membrane was blocked, immunoblotted with an appropriate antibody, treated with ECL solution (Amersham Pharmacia Biotech), and visualized in Amersham™ Imager 600 (GE Healthcare, Chicago, IL, USA).

(69) 4. Inhibitors and siRNA

(70) CDK4 inhibitor (PD0332991), JNK inhibitor (JNK-IN-8), MEK1 inhibitor (U0126), p38 MAPK inhibitor (SB203580) and rapamycin (R8781) were purchased from Sigma-Aldrich. Cells were treated with CDK4 inhibitor (500 nM), JNK inhibitor (1 μ M), MEK1 inhibitor (1 μ M), p38 MAPK inhibitor (1 μ M) or rapamycin (100 nM) and harvested at the designated time points after serum stimulation. Before serum starvation, a knockdown assay was performed by transfecting HEK293 cells with 50 nM siRNA using RNAiMAX (Invitrogen, CA, USA). Cells were harvested at the designated time points after serum stimulation. BRD2, p53, CDK4, RNF2 and Cyclin D1 siRNA were purchased from Bioneer (Daejeon, Korea). HDAC4 siRNA was purchased from Cell Signaling Technology. The sequence of siRNA is shown in Table 1 below.

(71) TABLE-US-00001

TABLE	1	Name	Sequence (5'→3')	SEQ. ID.	NO:
si-BRD2					
CACUUGGCCUGCAU	SEQ.	ID.	NO:	1	sense GACUA
UAGUCAUGCAGGCC	SEQ.	ID.	NO:	2	antisense AAGUG
si-p53					
CAGUUUGAGGUGCG	SEQ.	ID.	NO:	3	sense UGUU
si-p53					
AACACGCACCUCAA	SEQ.	ID.	NO:	4	antisense AGCUG
si-CDK4					
CCAGAAUCUACAGC	SEQ.	ID.	NO:	5	sense UACCA
si-CDK4					
UGGUAGCUGUAGAU	SEQ.	ID.	NO:	6	antisense UCCUGG
si-Cyclin D1					
GACCUUCGUUGCCC	SEQ.	ID.	NO:	7	sense UCUGU
si-Cyclin D1					
ACAGAGGGCAACGA	SEQ.	ID.	NO:	8	antisense AGGUC
si-RNF2					
UGAUAGGGUAUUGA	SEQ.	ID.	NO:	9	sense GUGUA
si-RNF2					
UACACUCAAUACCC	SEQ.	ID.	NO:	10	antisense UAUCA

5. Flow Cytometry

(72) Cells were harvested and processed using FITC-Annexin V Apoptosis Detection Kit I (BD Biosciences, San Jose, CA, USA) and propidium iodide DNA staining protocol. Apoptosis and cell cycle were analyzed by flow cytometry on a BD FACS caliber machine (BD Biosciences). All data were analyzed using FlowJo software (<https://www.flowjo.com>).

Experimental Example 1: Confirmation of SWI/SNF and TFIID Invoked by Runx3-BRD2 Nucleosome Complex

(73) Determination of the restriction point is made 3 to 4 hours after serum stimulation. It has been disclosed that Runx3-BRD2 complex is formed 1 to 2 hours after serum stimulation, and this complex contributes to the R-point determination by regulating hundreds of genes (Chi, X. Z. et al. Runx3 plays a critical role in restriction-point and defense against cellular transformation. Oncogene 36, 6884-6894 (2017).). BRD2 contains two bromodomains (BD1 and BD2), each of which interacts with a unique protein. The interaction between Runx3 protein and other proteins was confirmed as follows after mitogenic stimulation.

(74) As shown in FIG. 1a, BD1 was bound to the lysine residues 94 and 171 of the acetylated Runx3, BD2 was bound to the lysine residue 5 (H4K5-ac) of the acetylated histone 4, the lysine residue 12 (H4K12-ac) of histone 4 and the lysine residue 14 (H3K14-ac) of histone 3, and BRD2 was bound to SWI/SNF and TFIID through the C-terminal portion. In addition, RUNX3 was bound to these complexes through BRD2.

(75) In particular, as shown in FIG. 1b, not only the interactions between BRD2, RUNX3, and the lysine 12 of the acetylated histone 4, but also the interactions between p300, Runx3 and the lysine 12 of the acetylated histone 4 were observed 1 to 2 hours after mitogenic stimulation. The interaction between RUNX3 and the lysine residue 12 of the acetylated histone 4 was significantly reduced due to the knockdown of BRD2.

(76) The above results indicate that RUNX3 guided p300 to the target locus before the R-point, which acetylated histones, and that BRD2 was bound to both the acetylated RUNX3 and histones through two bromodomains.

(77) In addition, as shown in FIG. 1c, it was found that TAF1 (activated TAF1), TAF7 (suppressed TAF) and TBP formed a complex with BRD2 and RUNX3 1 hour after mitogenic stimulation. Thereafter, TAF7 was separated from the complex, and TFIID was activated after RUNX3-BRD2 binding. In addition, TAF1 and TBP were separated from RUNX3 4 hours after mitogenic

stimulation. Similarly, BRG-1 and BAF155 (components of SWI/SNF complex) were also bound to RUNX3 and BRD2 1 to 2 hours after mitogenic stimulation and separated 4 hours after mitogenic stimulation. The binding of the RUNX3-BRD2 complex to the SWI/SNF and TFIID complex was confirmed through proximity ligation analysis (PLA).

(78) The above results consistently suggest that the expression of the proteins related to R-point [p14ARF (hereinafter referred to as ARF), p53 and p21] was induced at the same time as when RUNX3 was bound to BRD2, SWI/SNF and TFIID.

(79) The BRD2 knockdown experiment showed that the SWI/SNF and TFIID complex was bound to RUNX3 through BRD2. The above results suggest that the transient formation of the RUNX3-BRD2 complex was specific to the R-point.

Experimental Example 2: Confirmation of Formation of Rpa-RX3-TR by Binding of CDK4 and Cyclin D1

(80) E2F1 was bound to RUNX3 1 to 2 hours after serum stimulation (FIG. 2a). When the 795.sup.th serine of pRB was phosphorylated by Cyclin D1-CDK4/6, the pRB-E2F1 complex was released from RUNX3 (FIG. 2a). These results suggest that CDK4 approached Rpa-RX3-AC to phosphorylate pRB. Immunoprecipitation/immunoblotting analysis showed that CDK4 was bound to RUNX3 after 2 hours, and that the binding was then weakened (FIG. 2a). It was confirmed through proximity ligation analysis (PLA) that this physical binding between RUNX3 and CDK4 was visible 2 hours after serum stimulation (FIG. 2b). These results indicate that CDK4 was a component of Rpa-RX3-AC.

(81) Although pRB and CDK4 were bound to Rpa-RX3-AC 2 hours after serum stimulation, the phosphorylation of pRB caused by CDK4 occurred 4 hours after stimulation (FIG. 2a). Through the analysis of the binding time between CDK4 and its binding proteins, it was confirmed that CDK4 was bound to p16 1 hour after stimulation, and then changed from p16 to p21 after 2 hours (FIG. 2a). P21 promotes the binding of Cyclin D1 to CDK4. However, Cyclin D1, which was expressed from 2 hours after stimulation, was bound to RNF2, but no CDK4 was bound to p21 at the same time (FIG. 2a). The Cyclin D1-CDK4 binding was observed after Rpa-RX3-TR was formed (4 hours after stimulation) (FIG. 2a). In particular, the knockdown of CDK4 significantly reduced the binding of RUNX3 to Cyclin D1 and HDAC4 (FIG. 2c). Therefore, the binding of RUNX3-BRD2 and the expression of ARF were maintained for up to 8 hours (FIG. 2c). These results suggest that CDK4 of Rpa-RX3-AC and Cyclin D1 of PRC1-Cyclin D1-HDAC4 provide docking sites for binding of two complexes capable of forming Rpa-RX3-TR.

Experimental Example 3: Confirmation of Various Pathways Contributing to R-Point Transition

(82) The binding of BRD2-RUNX3 and p300-RUNX3 was inhibited by the treatment of a MEK1 inhibitor (FIG. 3a). The above results indicate that the RAS-MEK signaling pathway stimulated the formation of Rpa-RX3-AC.

(83) The binding of Cyclin D1-CDK4 occurred 4 hours after serum stimulation, and the phosphorylation of pRB dependent on Cyclin D1-CDK4 also occurred at the same time (FIG. 2a). The knockdown of Cyclin D1 effectively reduced the phosphorylation of pRB (S795), but did not affect the phosphorylation of RUNX3 (S356) (FIG. 3b). These results suggest that CDK4 was activated independently of Cyclin D1 to promote the phosphorylation of RUNX3.

(84) The activation of CDK4 depends on the activation of CDK-activating kinase (CAK). JNK has been known as one of CAKs that phosphorylate T172 of CDK4. It was found that JNK phosphorylated T172 of CDK4 2 hours after serum stimulation (FIG. 3c). The pharmacological inhibition of JNK activity remarkably reduced the phosphorylation of the 356.sup.th serine of RUNX3 and maintained Rpa-RX3-AC for up to 8 hours (FIG. 3d). Similarly, the knockdown of JNK by siRNA also reduced the phosphorylation of the 356.sup.th serine of RUNX3 and maintained Rpa-RX3-AC (FIG. 3e). The over-expressed CDK4-WT phosphorylated Myc-RUNX3 2 to 4 hours after serum stimulation. However, Flag-CDK4-T172A (a mutant made to prevent phosphorylation by JNK) did not phosphorylate the 356.sup.th serine of RUNX3 (FIG. 3f). Flag-

CDK4-T172A was bound to RUNX3 even earlier after serum stimulation and did not separate for a long time (FIG. 3f). Consistently, Flag-CDK4-T172A continued to maintain Rpa-RX3-AC and also extended the expression of ARF up to 8 hours (FIG. 3f). These results suggest that the JNK pathway also contributed to the R-point transition through the activation of CDK4.

(85) It is known that the transcription and translation of Cyclin D1, which plays an important role in the formation of Rpa-RX3-TR, is stimulated through the respective RAS-RAF and RAS-PI3K pathways. As expected, the knockdown of a PI3K catalytic subunit (PIK3CA, encoding p110 α) reduced the level of Cyclin D1, maintained Rpa-RX3-AC, and extended the expression of ARF up to 8 hours (FIG. 3g). The suppression of mTOR, a downstream effector of the RAS-PI3K-AKT pathway, using rapamycin, also maintained Rpa-RX3-AC and extended the expression of ARF up to 8 hours (FIG. 3h). These results suggest that the PI3K pathway also contributed to the conversion of Rpa-RX3-AC to Rpa-RX3-TR by inducing Cyclin D1.

(86) The above results suggest that the three major pathways downstream of RAS (MEK, JNK, and PI3K) contributed to the R-point conversion at a distinct stage. The contribution at each step to the R-point of these pathways is summarized in FIG. 4.

Experimental Example 4: Confirmation of Effect of Improving Stability of Runx3-Brd2 Complex by Treatment of CDK4 Inhibitor and mTOR Inhibitor Compounds in Cells

(87) In Experimental Examples 2 and 3, the main action points in the process of forming the Rpa-Rx3-TR complex from the Rpa-Rx3-AC complex and then converting to the Rpa-Rx3-RE complex were the phosphorylation of Runx3 by CDK4, the binding of the phosphorylated Runx3 to cyclinD1, the PI3K-AKT-mTOR pathway, which activates cyclinD1, and the RAC-JNK pathway, another RAS pathway. Therefore, it was confirmed that the Rpa-Rx3-AC complex could be sustained by the treatment of a CDK4 inhibitor, mTOR inhibitor, or siRNA to inhibit protein synthesis of major pathways of action.

(88) Accordingly, the present inventors predicted that the treatment of a CDK4 inhibitor (PD0332991) would increase the effect of sustaining the physical binding between Runx3 and BRD2, and performed the following experiment.

(89) Particularly, the physical binding between Runx3 and Brd2 1 to 8 hours after serum stimulation was confirmed in an experiment treated with a CDK4 inhibitor at the concentration of 1 nM or more. In experiments not treated with a CDK4 inhibitor or treated with a CDK4 inhibitor at the concentration of 1 nM, the physical binding between Runx3 and Brd2 was not observed 8 hours after serum stimulation.

(90) As a result, as shown in FIG. 5, it was confirmed that the binding of Runx3 and BRD2 was maintained for up to 8 hours when the CDK4 inhibitor (PD0332991) was present at the concentration of 11 nM known to be non-toxic to normal cells.

(91) As shown in FIG. 6, it was also confirmed that the binding of Runx3 and BRD2 was maintained for up to 8 hours when the mTOR inhibitor (rapamycin) was present at the concentration of 100 nM known to be non-cytotoxic to normal cells.

Experimental Example 5: Confirmation of Cancer Apoptotic Effect by Co-Treatment of Runx3 Protein and CDK4 Inhibitor or mTOR Inhibitor Compound

(92) In Experimental Example 4, it was confirmed that the CDK4 inhibitor or the mTOR inhibitor compound increased the time for maintaining the Runx3-Brd2 complex. It was also confirmed that the cancer apoptotic effect was significantly increased when the CDK4 inhibitor or mTOR inhibitor was administered simultaneously with Runx3 than when Runx3 was administered alone.

(93) Particularly, each cell line prepared in Experimental Methods 1 and 4 was harvested and processed using FITC-Annexin V Apoptosis Detection Kit I (BD Biosciences, San Jose, CA, USA) and propidium iodide DNA staining protocol. Apoptosis and cell cycle were analyzed by flow cytometry on a BD FACS caliber machine (BD Biosciences). All data were analyzed using FlowJo software (<https://www.flowjo.com>).

(94) As a result, as shown in FIGS. 7a to 7c, it was confirmed that the cancer apoptotic effect when

Runx3 was introduced into Runx3 deficient cancer cells and when Runx3 and the CDK4 inhibitor (PD0332991) were simultaneously introduced in the presence of the CDK4 inhibitor at the concentration of 11 nM known to be non-toxic to normal cells, and the cancer apoptotic effect when Runx3 was introduced into Runx3 deficient cancer cells and when Runx3 and the CDK4 inhibitor were simultaneously introduced in the presence of the mTOR inhibitor (rapamycin) at the concentration of 100 nM known to be non-cytotoxic to normal cells.

(95) As shown in FIG. 8, it was also confirmed that the cancer apoptotic effect was significantly increased in H460 lung cancer cell line, MKN28 stomach cancer cell line and PANC1 pancreatic cancer cell line when the adenovirus expressing Runx3 protein and the CDK4 inhibitor or the mTOR inhibitor were administered simultaneously than when the adenovirus expressing Runx3 protein was administered alone. The above results suggest that the CDK4 inhibitor or mTOR inhibitor can be effectively used for treating cancer by administering the same in combination with Runx3.

Claims

1. A method for treating K-Ras mutant non-small cell lung cancer in a subject in need thereof, comprising: administering a viral vector comprising a nucleic acid molecule encoding Runx3 (Runt-related transcription factor 3) at a therapeutically effective apoptotic amount for treating K-Ras non-small cell lung cancer, wherein the Runx3 comprises the amino acid sequence of SEQ ID NO: 11 or 12, in combination with administering a non-therapeutic, non-apoptotic amount of palbociclib at a concentration of 11 nM; wherein the therapeutically effective apoptotic amount of Runx3 leads to apoptosis of mutant K-Ras non-small cell lung cancer cells, and wherein said apoptosis is increased by the combination of the non-therapeutic, non-apoptotic amount of palbociclib at a concentration of 11 nM.
 2. A method for treating K-Ras mutant non-small cell lung cancer in a subject in need thereof, comprising: administering a viral vector comprising a nucleic acid molecule encoding Runx3 (Runt-related transcription factor 3) at a therapeutically effective apoptotic amount for treating K-Ras non-small cell lung cancer, wherein the Runx3 comprises the amino acid sequence of SEQ ID NO: 11 or 12, in combination with administering a non-therapeutic, non-apoptotic amount of rapamycin at a concentration of 100 nM; wherein the therapeutically effective apoptotic amount of Runx3 leads to apoptosis of mutant K-Ras non-small cell lung cancer cells, and wherein said apoptosis is increased by the combination of the non-therapeutic, non-apoptotic amount of rapamycin at a concentration of 100 nM.
 3. The method for treating lung cancer according to claim 1, wherein the viral vector is selected from a group consisting of retrovirus, adenovirus, herpes simplex virus and lentivirus.
 4. The method for treating lung cancer according to claim 2, wherein the viral vector is selected from a group consisting of retrovirus, adenovirus, herpes simplex virus and lentivirus.
 5. The method for treating lung cancer according to claim 1, wherein the viral vector is an adenovirus.
 6. The method for treating lung cancer according to claim 2, wherein the viral vector is an adenovirus.
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